

Shervin Naseri

Senior Machine Learning Engineer | Backend Developer

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PROFESSIONAL SUMMARY

Machine Learning Engineer and Backend Developer with a strong background in Computer Vision and NLP. Committed to building reliable AI pipelines and optimizing deep learning models for performance. Experienced in full-cycle development, from researching novel architectures to deploying scalable solutions in production environments.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C++, Bash/Shell, SQL
Machine Learning: PyTorch, TensorFlow, Hugging Face, Transformers, RoPE, PEFT, RAG (LangChain/Weaviate), OpenCV, spaCy, JAX
Cloud & Backend: Node.js, Express, MongoDB, Elasticsearch, Redis, Docker, Nginx, AWS (EC2, S3), Git, CI/CD, Spark
Certifications: AWS Certified Generative AI - Professional

PROFESSIONAL EXPERIENCE

AI & Software Engineering Consultant

Independent Contracts | Ottawa, ON

March 2025 - Present

- Architected custom RAG pipelines and language model integrations for startup clients, collaborating with technical stakeholders to deploy scalable semantic retrieval solutions.
- Streamlined data engineering workflows and optimized PyTorch pipelines, resolving data throughput and memory bottlenecks for model fine-tuning and inference.
- Audited legacy Node.js and Python microservices to resolve application bottlenecks, ensuring stable backend support for predictive feature deployment.

Senior ML Engineer

Pardazesh Etelaat Kian (IraPardaz) | Tehran, Iran

June 2020 - Sep 2021

- Productionized real-time vehicle detection pipelines utilizing YOLOv4 architectures with custom regression logic for perspective distortion.
- Developed a production-grade Farsi OCR engine with specialized algorithms for complex paragraph block and poetry indentation.
- Designed multi-object tracking algorithms for lane-specific vehicle counting and municipal traffic analysis.
- Built a perceptual similarity search engine for a real-estate platform to identify duplicate listings and group visual assets.
- Led technical interviews and mentored junior data scientists on best practices in model deployment.

Core Backend Engineer

Nikaat Corp | Tehran, Iran

Dec 2016 - April 2020

- Architected and maintained scalable RESTful services for 'Mint' and 'Phosphor' product portfolios using Node.js, Express, and MongoDB.
- Designed and deployed a Bi-LSTM Seq2Seq architecture for domain-specific currency exchange rate forecasting.
- Engineered a GRU-based encoder-decoder model to predict complex women's health metrics from sparse, multi-modal data.
- Managed CI/CD pipelines and containerized production environments using Docker and Nginx.

Python Developer

Vestaak | Tehran, Iran

March 2016 - Dec 2016

- Developed a content recommendation system using Scikit-learn and user interaction data mined from Telegram API.

SELECTED PROJECTS

CodebaseMCP (RAG for Code Analysis) | [Github](#)

- Engineered a Model Context Protocol (MCP) server enabling LLMs to semantically query and index local environments.
- Built a hybrid retrieval pipeline using Weaviate and custom Python AST parsing for precision function signature indexing.

EZTrainer (Deep Learning Optimization) | [Github](#)

- Developed a modular PyTorch training framework to benchmark novel regularization techniques against standard baselines.
- Implemented Adaptive Batch Normalization layers to improve convergence on non-i.i.d. datasets.

Temporal-Integral Diffusion Adapter (PEFT Framework) | [Github](#)

- Developed a PyTorch PEFT framework implementing fractional positional encodings via patched RoPE.
- Engineered a training pipeline using staircase attention masking for adaptive micro-step computation.

EDUCATION

MSc. Computer Science

University of Ottawa | Ottawa, ON

Sep 2021 - March 2025

- Research Focus: Simulation Reliability in Autonomous Systems. Co-authored 'Evaluating the impact of flaky simulators on testing autonomous driving systems' (Accepted at EMSE2024).

MSc. Software Engineering

Sharif University of Technology | Tehran, Iran

Graduated

- Thesis: Graph Similarity Estimation on large-scale networks using sampling algorithms applied to Protein Interaction data.

BSc. Computer Science

Amirkabir University of Technology | Tehran, Iran

Graduated

- Publication: 'Counting Closed Billiard Paths' (Accepted at ICCG2019).